

# YIFAN ZHU

Northwestern Polytechnical University · Software Engineering · 2023 Undergraduate  
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## EDUCATION

Northwestern Polytechnical University, Software Engineering, *B.Eng. Candidate* 2023.09 – Present

- **Academic Performance:** GPA 3.813/4.1, ranked 11/270; awarded the National Scholarship for Undergraduate Students for two consecutive years.
- **English Proficiency:** CET-4 613; CET-6 587; CET-6 Speaking Test: Excellent.

## RESEARCH EXPERIENCE

### BrepLLM: Enabling Large Language Models to Understand Boundary Representations

ECCV 2026 Accepted, *Fifth Author*

- Worked on B-rep data understanding in the CAD domain, exploring how LLMs can directly understand native geometric and topological structures without relying on point clouds, images, meshes, or modeling command sequences.
- Mainly contributed to paper writing, revision, formatting, and technical route summarization; gained experience in multimodal model architectures, structured data representation, and academic writing.

### Unified Multimodal Agent Framework for CadQuery-based 3D-CAD Generation, Understanding and Editing

Ongoing

- Participating in a unified multimodal agent framework for 3D-CAD generation, understanding, and editing, using executable Python CadQuery programs as a unified modeling representation.
- Conducted research and experiments on CadQuery data construction, unified SFT data format design, model fine-tuning, and CAD-kernel-based automated verification and repair.

## INTERNSHIP EXPERIENCE

Shenzhen Kuakua Jingling Technology Co., Ltd., Backend Development Intern 2025.07 – 2025.08

- Completed workflow orchestration for a digital employee Agent, enabling automated enterprise document generation and participating in real-world AI Agent system implementation.
- Assisted in backend API development for a municipal big data bureau project, covering requirement understanding, interface implementation, integration testing, and delivery.
- Conducted research on AI multi-turn dialogue QA and knowledge-base products, and wrote a research report for technology selection and product planning.

## PROJECT EXPERIENCE

### Dual-modal UAV Urban Traffic Perception and Intelligent Analysis System

National-level Undergraduate Innovation Training Project, completed with a Good rating

- Built a UAV-based urban traffic object detection and visualization system using visible-light and infrared dual-modal fusion for multiple urban scenarios and traffic object categories.
- Participated in YOLOv8-OBDD dual-modal model integration and deployment; designed an Edge-Cloud-Dashboard architecture for real-time reporting, forwarding, and visualization of detection results and video streams.

### AgriGuard: Smart Agricultural Pest Control Management Platform

Agile Software Development Course Project

- Built an end-to-end digital pest control system for agricultural scenarios, integrating pest image recognition, RAG-based control suggestions, task scheduling, work-log tracking, and risk prediction.
- Worked in a Scrum team and completed three Sprint iterations; responsible for backend APIs of the pest recognition module, YOLO model training and deployment, and part of the frontend implementation.

## SELECTED HONORS AND AWARDS

- Awarded the **National Scholarship for Undergraduate Students** twice, December 2024 and December 2025.
- 2026 Mathematical Contest in Modeling (MCM), **Honorable Mention**, May 2026.
- 2025 Contemporary Undergraduate Mathematical Contest in Modeling, **Shaanxi Second Prize**, November 2025.
- 18th National Undergraduate Software Innovation Competition, **Northwest Regional Third Prize**, April 2025.
- 16th National College Student Mathematics Competition; 2025 11th National College Students Statistical Modeling Competition, **Provincial / Shaanxi Third Prize**.